



## Lecture # 7

# Project Selection and Project Charter

## ENGINEERING PROJECT MANAGEMENT

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# Project selection

- ▶ Strategic planning
- ▶ Methods for selecting a project

# Strategic planning

- ▶ Strategic planning involves determining long-term objectives, predicting future trends, and projecting the need for new products and services
- ▶ SP technique - Organizations often perform a **SWOT analysis**
- ▶ Analyzing **S**trengths, **W**eaknesses, **O**pportunities, and **T**hreats
- ▶ As part of strategic planning, organizations
  - ▶ Identify potential projects
  - ▶ Use realistic methods to select which projects to work on

# Methods for selecting a project

- ▶ Problem: There are usually more projects than available time and resources to implement them
- ▶ Methods for selecting projects include:
  - ▶ (1) focusing on broad organizational needs
  - ▶ (2) categorizing projects (Discipline)
  - ▶ (3) performing net present value or other financial analyses
  - ▶ (4) using a weighted scoring model
  - ▶ (5) implementing a balanced scorecard

# Financial considerations

- ▶ Financial considerations are often an important consideration in selecting projects
- ▶ Three primary methods for determining the projected financial value of projects:
  - ▶ Net present value (NPV) analysis
  - ▶ Return on investment (ROI)
  - ▶ Payback analysis

# Weighted Score Model

| Weighted Decision Matrix |           |           |       |           |       |           |       |
|--------------------------|-----------|-----------|-------|-----------|-------|-----------|-------|
|                          |           | OPTIONS   |       |           |       |           |       |
| Criteria                 | Weighting | Company A |       | Company B |       | Company C |       |
|                          |           | Score     | Total | Score     | Total | Score     | Total |
| Cost                     | 5         | 5         | 25    | 3         | 15    | 4         | 20    |
| Service Level            | 4         | 3         | 12    | 5         | 20    | 2         | 8     |
| Ease of Termination      | 4         | 2         | 8     | 5         | 20    | 5         | 20    |
| Contract Length          | 2         | 4         | 8     | 4         | 8     | 3         | 6     |
| Financial Strength       | 3         | 4         | 12    | 5         | 15    | 3         | 9     |
|                          | TOTAL:    |           | 65    |           | 78    |           | 63    |

# Balanced Scorecard



# Project Charter

- ▶ A project charter is **a short, straightforward document that serves as the foundation for a project**. The project charter functions as both the project's internal marketing tool and reference guide



# Project Charter - Focus

